

```
identifier [ id-qualifier ] : type-or-ref // named alternate
```

```
id-qualifier:
  tag
```

CHOICE OF declares an element that MAY be any of a set of TLV types. **CHOICE OF** is considered a **pseudo-type**, rather than a fundamental type, in that the **CHOICE OF** itself doesn't have a representation in the final TLV encoding.

The allowed TLV types for a **CHOICE OF**, known as **alternates**, are given in the body of the definition. An **alternate** MAY be any of the fundamental TLV types, an **ANY** pseudo-type, or another **CHOICE OF** definition (more on this below). Additionally, an alternate MAY be a type reference (in the form of a scoped type name) referring to a type defined outside of the **CHOICE OF** definition.

A **nullable** qualifier MAY be used to indicate that a TLV **Null** can be encoded in place of the **CHOICE OF**. This is exactly the same as if NULL had been listed as one of the alternates.

B.4.2.1. Alternate Names and Tags

Alternates MAY be assigned textual names to distinguish them from one another. Each such name SHALL be unique within the particular **CHOICE OF** definition. Alternate names do not affect the encoding of the resultant TLV. Rather, alternate names serve as user documentation, or as input to code generation tools.

Named **CHOICE OF** alternates MAY include a **tag** qualifier assigning a particular **tag** value to the **alternate**. When qualified in this way, the given tag value serves as a default tag for the alternate whenever the **CHOICE OF** appears in a context that doesn't otherwise specify a tag. The tags assigned within a **CHOICE OF** do not need to be unique, although see the discussion of **Ambiguous Alternates** below.

Both protocol-specific and context-specific tags are allowed on the alternates of a **CHOICE OF** definition.

B.4.2.2. Nested CHOICE OF and CHOICE OF Merging

It is legal for an alternate within a **CHOICE OF** to be another **CHOICE OF** definition, or a type reference to such. In this case, the effect is exactly as if the alternates of the inner **CHOICE OF** definition had been declared directly with the outer definition. This merging of **CHOICE OF** alternates occurs to any level of nesting, and MAY be used as a means of declaring multiple **CHOICE OF** that are supersets of other **CHOICE OF**.

When alternates are merged, their names are preserved. In cases where the same name appears in nested **CHOICE OF** definitions, the name of the outer alternate is prepended to that of the inner alternate, separated by a dot, to form a unique name for the merged alternate. In these cases, the outer alternate SHALL have a name in the schema, to ensure uniqueness.

An example of invalid **CHOICE OF** syntax, which results in a name conflict when alternates are merged: